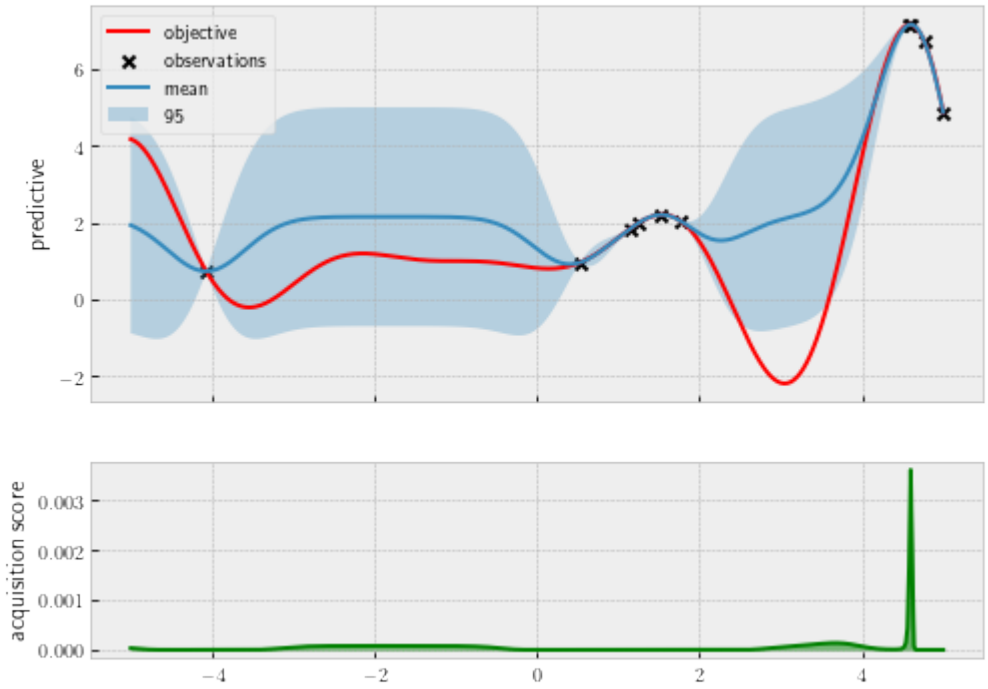


Figure 1.13 Bayesian optimization has found the global optimum at the end of the search.

This example has clearly shown us that Bayesian optimization can work a lot better than random search and grid search. This should be a very encouraging sign for us, considering that the latter two strategies are what many machine learning practitioners are using when faced with a hyperparameter tuning problem.

For example, Scikit-learn is one of the most popular machine learning libraries in Python, which offers the `model_selection` module for various model selection tasks, including hyperparameter tuning. However, random search and grid search are the only hyperparameter tuning methods implemented in the module. In other words, if you are indeed tuning your hyperparameters with random or grid search, there is a lot of headroom for you to do better.

Overall, employing Bayesian optimization may result in a drastic improvement in optimization performance. We can take a quick look at a couple of real-world examples.

- This research paper (https://arxiv.org/pdf/2104.10201.pdf), which was the result from a joint study by Facebook, Twitter, Intel, and others, found that Bayesian optimization was extremely successful across many hyperparameter tuning tasks.
- Frances Arnold, Nobel prize winner in 2018 and professor at Caltech, uses Bayesian optimization in her research to guide the search for enzymes that are efficient at catalyzing desirable chemical reactions.
- This study (https://www.nature.com/articles/nmat4717) that was published in Nature applied Bayesian optimization to the problem of screening for molecular organic light-emitting diodes (an important type of molecules) and observed a big improvement in efficiency.

And there are many more of these examples out there.

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